

A14 MICROBE AND MOLD CONTROL | O (MAX : 1 PT)

Intent : Reduce mold and bacteria growth within the building mechanical system.

Summary : This WELL feature requires projects to utilize UVGI systems and/or conduct regular inspections on components of the cooling system to reduce or eliminate growth of microbes and mold.

Issue : Mold will grow in places with an acceptable temperature range, a nutrient source and sufficient moisture.¹ Air conditioner systems, specifically cooling coils, have high levels of moisture condensation and serve as the perfect breeding grounds for mold.² Mold developed on cooling coils may shed particles into the building's indoor air and trigger asthma, headaches, allergies and other respiratory system disorders.³⁻⁶ Exposure to mold has also been associated with hypersensitivity pneumonitis, allergic rhinitis, bronchitis, lung tumor development, eczema and toxic mold syndrome.^{5,7}

Solutions : The growth of microorganisms and mold can be prevented and mitigated through a combination of regular mold inspections and placement of microbe inactivation techniques, such as Ultraviolet Germicidal Irradiation (UVGI) systems. Periodic inspections and maintenance of cooling systems are good preventative methods to reduce system failures that contribute to an increased potential for mold proliferation. Expenses associated with proper maintenance will often offset typical costs associated with mold remediation.² UVGI has been shown to dramatically reduce mold and bacteria growth on cooling coils and destroy microbial films that accumulate on their surfaces.⁸ Studies have also associated the implementation of UVGI systems with a simultaneous reduction in viable microorganisms and respiratory disorders in the workplace.⁹

PART 1 IMPLEMENT ULTRAVIOLET TREATMENT FOR HVAC SURFACES (MAX : 1 PT)

For All Spaces:

1: UV system design

The following requirements are met:

- a. All central air handling units use ultraviolet lamps to irradiate the surfaces of the cooling coils and drain pans.¹⁰
- b. All cooling coils and drain pans not associated with central air handling units (e.g., those used in fan coil units or supplementary air handling units) either:
 1. Are irradiated by ultraviolet lamps.
 2. May be opened for inspection for mold growth and cleaned, if necessary.

2: UV system maintenance and inspection

The following requirements are met:

- a. Evidence that the ultraviolet lamps have been replaced or maintained, according to manufacturer's recommendation is submitted annually through the WELL digital platform.
- b. All cooling coils without ultraviolet lamps (if applicable) are inspected on a quarterly basis for mold growth and cleaned if necessary. Dated photos demonstrating adherence are submitted annually through the WELL digital platform.

REFERENCES

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APPENDIX A1:

$$eACH = Q / [\text{room volume}]^{18}$$

Where Q = A + B + C + D (total clean volumetric air flow rate)

To determine the total clean volumetric air flow rate (Q) for a room, sum the air flow rates for individual systems (as applicable to the HVAC system) per the calculations below.

A: Ventilation Systems¹⁸

The greater of:

The full mechanical HVAC outdoor air supply rate

The full mechanical HVAC system's exhaust rate, provided the transfer air is from non-regularly occupied spaces or outdoors

B: CADR-rated Media Filters (Standalone Systems)^{17,18}

One of the following:

The full m-CADR rate (AHAM Certified per ANSI/AHAM Standard AC-5) or, if unavailable,

The full smoke-CADR rate (AHAM Certified per ANSI/AHAM Standard AC-1)

C: In-duct or non CADR-rated Media Filters¹⁸

Multiply the air supply rate (e.g., mechanical HVAC fan, blower fan) by the weighted clean air below:

MERV-A (ASHRAE 52.2)	ePM2.5 (ISO 16890)	Weighted Clean Air Factor (MERV E2 [1-3 um] minimum arrestance efficiency)
< 13	≤ 79	0
13	80	85
14-15	85	90
16	90	95
17-19 (HEPA)	99	99

D: In-duct UVGI¹⁷

Multiply the air supply rate by the infectious aerosol reduction efficiency determined in accordance with ANSI/ASHRAE 185.1 with MS2 challenge organism.